

## PRISMA 2020 checklist

Quantum Computing for Medical Imaging Applications: A PRISMA-Compliant Systematic Review (IEEE Transactions on Radiation and Plasma Medical Sciences, submitted manuscript)

Locations give the section and the page of the submitted manuscript. "Repository" refers to <https://github.com/lpapp-muw/qc-medimaging-review>, folder docs/supplementary for Supplementary Tables S1 and S2.

| Section and topic       | Item | Checklist item                                                                                                                                                                                                                                                                                       | Location where item is reported                                                                                                                                                                                                                                                                                     |
|-------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>TITLE</b>            |      |                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                     |
| Title                   | 1    | Identify the report as a systematic review.                                                                                                                                                                                                                                                          | Title (p. 1).                                                                                                                                                                                                                                                                                                       |
| <b>ABSTRACT</b>         |      |                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                     |
| Abstract                | 2    | See the PRISMA 2020 for Abstracts checklist.                                                                                                                                                                                                                                                         | Abstract (p. 1). Unstructured abstract in the journal format. The PRISMA 2020 for Abstracts checklist was not completed separately.                                                                                                                                                                                 |
| <b>INTRODUCTION</b>     |      |                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                     |
| Rationale               | 3    | Describe the rationale for the review in the context of existing knowledge.                                                                                                                                                                                                                          | Section I (p. 1).                                                                                                                                                                                                                                                                                                   |
| Objectives              | 4    | Provide an explicit statement of the objective(s) or question(s) the review addresses.                                                                                                                                                                                                               | Section I, research questions RQ1 to RQ7 and SRQ1 to SRQ2 (pp. 1 to 2).                                                                                                                                                                                                                                             |
| <b>METHODS</b>          |      |                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                     |
| Eligibility criteria    | 5    | Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.                                                                                                                                                                                          | Criteria: Section II-B (p. 2). Grouping for the syntheses: Section II-G (p. 4) and Section II-J (p. 5), Table I (p. 5).                                                                                                                                                                                             |
| Information sources     | 6    | Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.                                                                                            | Section II-C (p. 2): PubMed, Web of Science Core Collection and IEEE Xplore, last searched on 18 May 2026 (pilot pass on 30 January 2026). No other sources were used.                                                                                                                                              |
| Search strategy         | 7    | Present the full search strategies for all databases, registers and websites, including any filters and limits used.                                                                                                                                                                                 | Section II-C (p. 2): full query string for each database, and the open-access filter of the pilot pass. Date and language limits: Section II-B (p. 2).                                                                                                                                                              |
| Selection process       | 8    | Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.                     | Section II-D (p. 3): title and abstract classification by an AI model (Claude Opus 4.7) with quote checking, the ionising sub-screen, and full-text eligibility decided by a deterministic rule, individual re-reading confirmed by the corresponding author, and co-author review. Use of AI: Section II-K (p. 6). |
| Data collection process | 9    | Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process. | AI extraction in five passes with quote checking: Section II-E (p. 3). Supplementary-material enrichment: Section II-F (p. 3). Independent co-author extraction of a validation subset: Section II-I (p. 5). Authors were not contacted: Section II-J (p. 5).                                                       |
| Data items              | 10a  | List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.                        | Extracted fields: Section II-E (p. 3). Quality and maturity measures: Section II-H (p. 4). Rule for selecting among several reported metric values: Section II-J (p. 5).                                                                                                                                            |

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|                               | 10b  | List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.                                                      | Variables: Section II-E (p. 3) and Section II-G (p. 4). Not-reported items were treated as missing: Section II-J (p. 5).                                                               |
| Study risk of bias assessment | 11   | Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process. | Section II-H (p. 4): custom list of deterministic flags derived from the AI extraction, because standard tools do not apply. Agreement with co-author extraction: Section II-I (p. 5). |
| Effect measures               | 12   | Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.                                                                                                                               | Paired quantum-versus-classical differences on the same metric, dataset and split: Section II-J (p. 5), reported in Section III-J (p. 14). No pooled effect measure.                   |
| Synthesis methods             | 13a  | Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).                                              | Pairing criteria and sensitivity criteria: Section II-J (p. 5) and Section III-H (p. 12).                                                                                              |
|                               | 13b  | Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.                                                                                                             | Parsing and selection of reported metric values, and missing data: Section II-J (p. 5). Derived fields: Section II-G (p. 4).                                                           |
|                               | 13c  | Describe any methods used to tabulate or visually display results of individual studies and syntheses.                                                                                                                                                            | Section II-J (p. 5). Tables and figures in Section III, per-study values in Supplementary Table S1 (repository).                                                                       |
|                               | 13d  | Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.       | Section II-J (p. 5): narrative synthesis with descriptive statistics. No meta-analysis, because of heterogeneity and the absence of a pooled effect estimate.                          |
|                               | 13e  | Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).                                                                                                                              | Section II-J (p. 5): cross-tabulations by paradigm, task, modality and execution context. No statistical heterogeneity analysis (no meta-analysis).                                    |
|                               | 13f  | Describe any sensitivity analyses conducted to assess robustness of the synthesized results.                                                                                                                                                                      | Pre-registered restriction to studies with a real clinical dataset, a classical baseline and statistical testing: Section II-J (p. 5) and Section III-H (p. 12).                       |
| Reporting bias assessment     | 14   | Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).                                                                                                                                           | Not performed. Stated in Section V (p. 19).                                                                                                                                            |
| Certainty assessment          | 15   | Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.                                                                                                                                                             | Not performed, with the reason. Stated in Section II-H (p. 4).                                                                                                                         |
| <b>RESULTS</b>                |      |                                                                                                                                                                                                                                                                   |                                                                                                                                                                                        |
| Study selection               | 16a  | Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.                                                                      | Section III-A (p. 6), Fig. 1 (p. 8).                                                                                                                                                   |
|                               | 16b  | Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.                                                                                                                                       | Section III-A (p. 6): reasons by category. Each of the 48 reports with its reason, route and recorded note: Supplementary Table S2 (repository).                                       |

| Section and topic             | Item | Checklist item                                                                                                                                                                                                                                                                       | Location where item is reported                                                                                                                                                                                                                                                                                                                                                                                     |
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| Study characteristics         | 17   | Cite each included study and present its characteristics.                                                                                                                                                                                                                            | Included studies cited in Section II-F (p. 3) and listed in the references. Characteristics: Section III-B (p. 6), Table III (p. 9), and per study in Supplementary Table S1 (repository).                                                                                                                                                                                                                          |
| Risk of bias in studies       | 18   | Present assessments of risk of bias for each included study.                                                                                                                                                                                                                         | Summaries: Section III-E (p. 10), Section III-F (p. 11), Table VI (p. 11). Per study: Supplementary Table S1 (repository). Seven-study subset: Table IX (p. 13) and Table X (p. 13).                                                                                                                                                                                                                                |
| Results of individual studies | 19   | For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.                                                     | Per-study characteristics: Supplementary Table S1 (repository). Reported quantum and classical metric values per study: extraction workbook in the repository (performance-metric fields). Per-study effect estimates with precision are not tabulated, and the per-paper records behind the paired comparison are not deposited (repository README). The paired comparison is summarised in Section III-J (p. 14). |
| Results of syntheses          | 20a  | For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.                                                                                                                                                                               | Paired comparison (datasets and execution of the 97 contributing studies): Section III-J (p. 14). Quality-fit subset: Section III-H (p. 12).                                                                                                                                                                                                                                                                        |
|                               | 20b  | Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect. | Descriptive syntheses: Section III-B (p. 6) to Section III-J (p. 14). Direction of the quantum-versus-classical differences: Section III-J (p. 14). No meta-analysis.                                                                                                                                                                                                                                               |
|                               | 20c  | Present results of all investigations of possible causes of heterogeneity among study results.                                                                                                                                                                                       | Cross-tabulations: Section III-C (p. 7), Table IV (p. 10), Section III-D (p. 7), Table V (p. 10).                                                                                                                                                                                                                                                                                                                   |
|                               | 20d  | Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.                                                                                                                                                                           | Section III-H (p. 12), with the expert evaluation of that subset in Section III-I (p. 12).                                                                                                                                                                                                                                                                                                                          |
| Reporting biases              | 21   | Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.                                                                                                                                                              | Not assessed. Stated in Section V (p. 19).                                                                                                                                                                                                                                                                                                                                                                          |
| Certainty of evidence         | 22   | Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.                                                                                                                                                                                  | Not assessed. Stated in Section II-H (p. 4).                                                                                                                                                                                                                                                                                                                                                                        |
| <b>DISCUSSION</b>             |      |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Discussion                    | 23a  | Provide a general interpretation of the results in the context of other evidence.                                                                                                                                                                                                    | Section IV (p. 15), subsections IV-A to IV-D (from p. 15).                                                                                                                                                                                                                                                                                                                                                          |
|                               | 23b  | Discuss any limitations of the evidence included in the review.                                                                                                                                                                                                                      | Section IV-B (p. 15) and Section IV-C (p. 17).                                                                                                                                                                                                                                                                                                                                                                      |
|                               | 23c  | Discuss any limitations of the review processes used.                                                                                                                                                                                                                                | Section V (p. 19).                                                                                                                                                                                                                                                                                                                                                                                                  |
|                               | 23d  | Discuss implications of the results for practice, policy, and future research.                                                                                                                                                                                                       | Section IV-E (p. 18), Fig. 7 (p. 16), Section VI (p. 19).                                                                                                                                                                                                                                                                                                                                                           |
| <b>OTHER INFORMATION</b>      |      |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Registration and protocol     | 24a  | Provide registration information for the review, including register name and registration number, or state that the review was not registered.                                                                                                                                       | Section II-A (p. 2): OSF Registries, Generalized Systematic Review Registration Form, <a href="https://osf.io/xu6t4">osf.io/xu6t4</a> (reference [8]), deposited on 29 June 2026, retrospective for Stages 1 to 4.                                                                                                                                                                                                  |
|                               | 24b  | Indicate where the review protocol can be accessed, or state that a protocol was not prepared.                                                                                                                                                                                       | <a href="https://osf.io/xu6t4">osf.io/xu6t4</a> (reference [8], Section II-A (p. 2)). A copy of the registration text is in the repository (docs/OSF_protocol_submitted_2026-06-30.md).                                                                                                                                                                                                                             |

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|                                                | 24c  | Describe and explain any amendments to information provided at registration or in the protocol.                                                                                                                                            | Section II-A (p. 2) and Supplementary Note S1 (supplementary PDF). Deviations as executed: predatory-journal rule in Section II-B (p. 2), model change for the re-extraction in Section II-E (p. 3).                                                                                                                                                                                                                                                                                                                                                    |
| Support                                        | 25   | Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.                                                                                                              | Funding section (p. 19): no external funding, no sponsor role.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Competing interests                            | 26   | Declare any competing interests of review authors.                                                                                                                                                                                         | Competing interests section (p. 19), and the statement in the Acknowledgment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Availability of data, code and other materials | 27   | Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review. | Data and code availability section (p. 20):<br><a href="https://github.com/lpapp-muw/qc-medimaging-review">https://github.com/lpapp-muw/qc-medimaging-review</a> . Template form: 4-FullTextExtraction/extract_v1/extraction_human_blank.xlsx. Extracted data: 7-Manuscript/supp_enrich/out/extractions_ai_enriched_v134.xlsx. Analysis code: recompute_v133.py and the pipeline scripts. Other materials: agent configurations, verdict files, kappa inputs. The per-paper records behind the paired comparison are not deposited (repository README). |

Checklist items from: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;372:n71. doi: 10.1136/bmj.n71. Licensed under CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>).